

# AI • Machine Learning & Deep Learning

Artificial Intelligence

Machine Learning

Deep Learning

# What We Will Cover Today

## SECTION 1



### Artificial Intelligence

- History
- Branches
- Turing Test
- AI Types
- Ethics

## SECTION 2



### Machine Learning

- Types
- Algorithms
- Workflow
- Evaluation
- Tools

## SECTION 3



### Deep Learning

- Neural Networks
- Architectures
- Training
- Applications

SECTION ONE

01

# Artificial Intelligence

*Definition · History · Branches · Types · Ethics & Future*

# What is Artificial Intelligence?

*"The science and engineering of making intelligent machines, especially intelligent computer programs." — John McCarthy, 1956*



## Perceive

AI senses the world —  
vision, speech, text,  
sensor data



## Reason

Processes information,  
draws conclusions,  
solves problems



## Learn

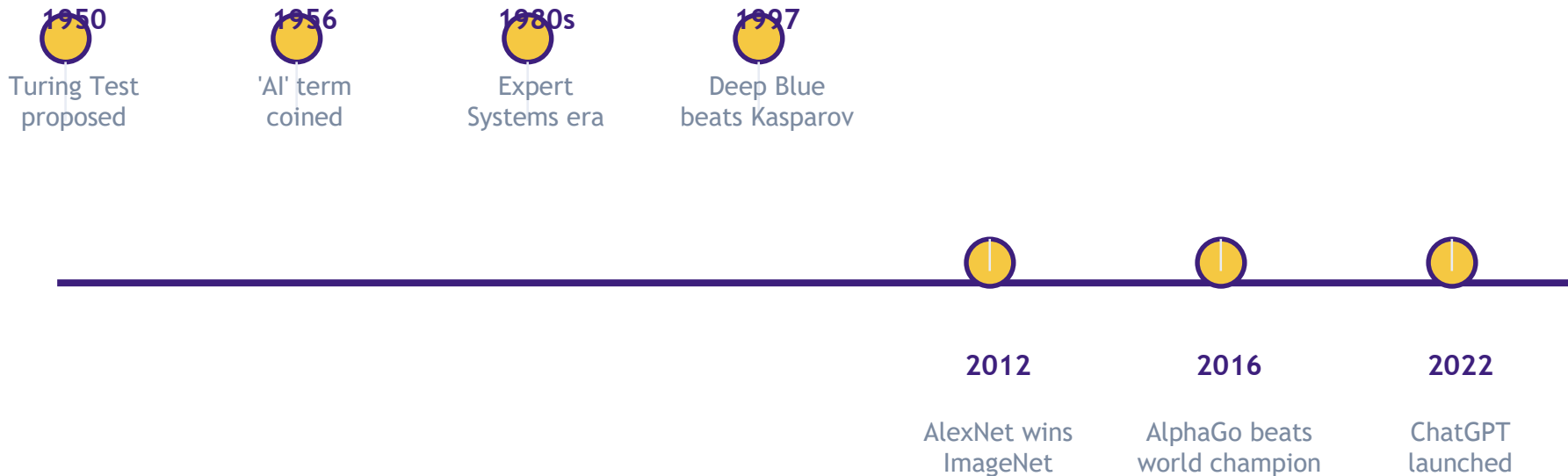
Improves performance  
from experience  
without reprogramming



## Act

Makes decisions,  
controls robots,  
generates outputs

# A Brief History of AI



✂ Key Insight: AI has gone through multiple "winters" (funding cuts) and "summers" (breakthroughs). We are now in the most exciting AI summer in history, driven by big data, powerful GPUs, and transformer architectures.

# Branches of Artificial Intelligence



## Machine Learning

Systems that learn from data



## Natural Language Processing

Understanding & generating human language



## Computer Vision

Interpreting images and video



## Robotics

Physical AI agents that act in the world



## Expert Systems

Rule-based decision making



## Generative AI

Creating text, images, audio, code

# Types of AI – By Capability



✓ EXISTS TODAY

## Narrow AI (ANI)

*Artificial Narrow Intelligence*

Excels at ONE specific task.  
Cannot generalise beyond its training.

### Examples:

Siri, Chess engines, Spam filters,  
Face recognition, Google Translate



🔬 RESEARCH GOAL

## General AI (AGI)

*Artificial General Intelligence*

Human-level reasoning across all domains. Can learn anything a human can.

### Examples:

Hypothetical – no AGI exists yet



🧠 HYPOTHETICAL

## Super AI (ASI)

*Artificial Superintelligence*

Far surpasses human intelligence in every field – science, creativity, social skills.

### Examples:

Hypothetical – depicted in sci-fi films

# AI Applications & Ethical Concerns

## Top AI Applications



### Healthcare

Diagnosis, drug discovery



### Autonomous Vehicles

Self-driving, navigation



### Virtual Assistants

Siri, Alexa, ChatGPT



### Cybersecurity

Threat detection, fraud



### Education

Personalised learning



### Manufacturing

Predictive maintenance

## Ethical Concerns



### Privacy

Mass surveillance and data collection



### Bias & Fairness

Discriminatory algorithmic decisions



### Job Displacement

Automation replacing human roles



### Autonomy & Control

Risks of autonomous AI systems



### Misinformation

Deepfakes, fake news generation



### Environmental Impact

Massive energy consumption of AI

SECTION TWO

02

# Machine Learning

*Types · Algorithms · Workflow · Evaluation · Tools*

# What is Machine Learning?

*"Machine Learning is a subset of AI that enables systems to automatically learn and improve from experience WITHOUT being explicitly programmed."*

## Traditional Programming

- You write rules manually
- Data + Rules  $\rightarrow$  Output
- Brittle: breaks on new data
- Hard to scale



## Machine Learning

- Algorithm learns rules
- Data + Output  $\rightarrow$  Rules (learned)
- Generalises to new data
- Scales with more data

# Three Types of Machine Learning



## Supervised Learning

### How it works:

Trained on labelled data  
(input+output pairs)

### Key Algorithms:

Linear Regression · Logistic  
Regression · SVM · Decision Tree ·  
Random Forest · KNN

### Applications:

*Email spam · House prices · Image  
classification · Medical diagnosis*



## Unsupervised Learning

### How it works:

Finds hidden patterns in unlabelled  
data

### Key Algorithms:

K-Means · DBSCAN · PCA ·  
Autoencoders · Apriori

### Applications:

*Customer segmentation · Anomaly  
detection · Recommendation · Topic  
modelling*



## Reinforcement Learning

### How it works:

Agent learns by trial & error with  
rewards/penalties

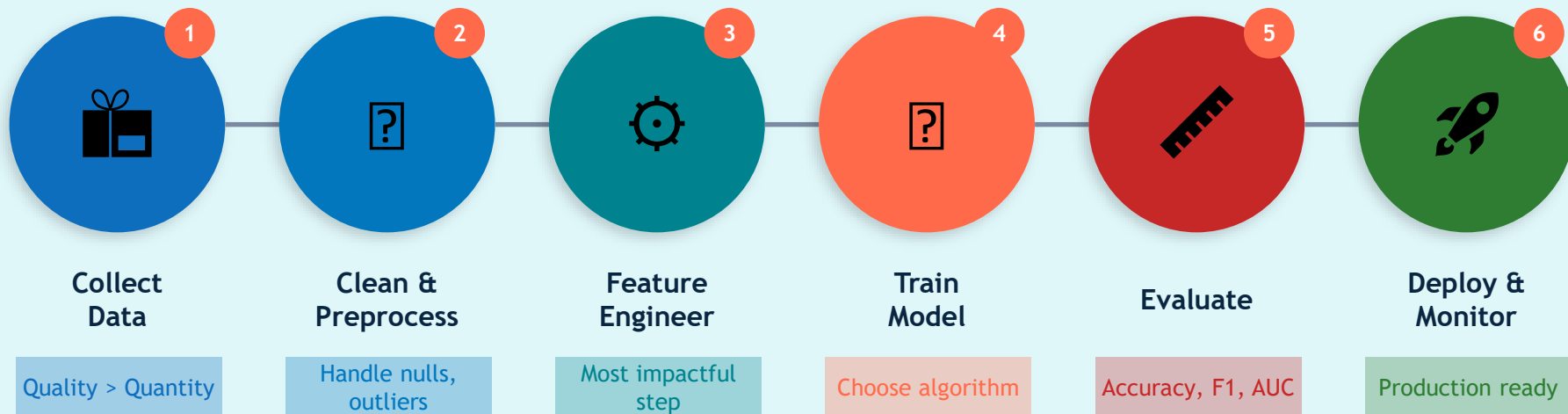
### Key Algorithms:

Q-Learning · SARSA · PPO · A3C · DQN

### Applications:

*Game AI · Self-driving · Robot control  
· Trading strategies*

# The Machine Learning Workflow



## Key Evaluation Metrics

**Accuracy:** % correct predictions overall

**F1 Score:** Harmonic mean of Precision & Recall

**Precision:** Of predicted positives, how many are right

**AUC-ROC:** Model's ability to distinguish classes

**Recall:** Of actual positives, how many found

# ML Ecosystem – Tools & Frameworks

## Python

Language

Industry standard for  
ML

## NumPy

Numerical

Array operations, linear  
algebra

## Pandas

Data

DataFrames, EDA

## Matplotlib

Visualise

Charts & plots

## Scikit-learn

Classical ML

100+ ML algorithms

## XGBoost

Boosting

Gradient boosting

## Jupyter

Notebook

Interactive  
development

## MLflow

MLOps

Track experiments &  
models

SECTION THREE

03

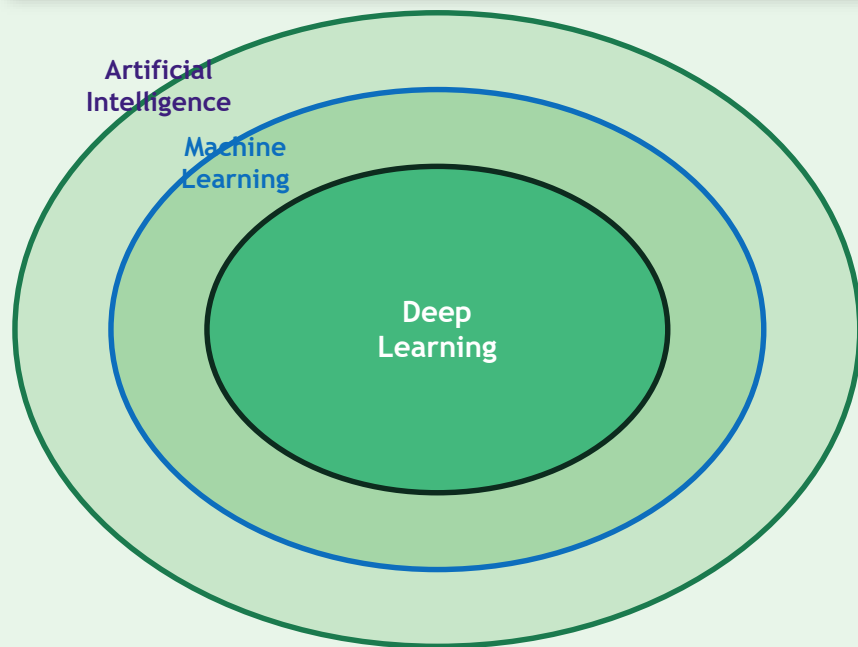
Deep

Learning

*Neural Networks · Architectures · Training · State of the Art*

# What is Deep Learning?

*"Deep Learning uses multi-layered artificial neural networks inspired by the human brain to learn representations directly from raw data – no manual feature engineering needed."*



| Feature             | Classical ML  | Deep Learning         |
|---------------------|---------------|-----------------------|
| Data needed         | Small–Medium  | Very Large (millions) |
| Feature Engineering | Manual        | Automatic             |
| Hardware            | CPU fine      | GPU / TPU required    |
| Interpretability    | High          | Low (black box)       |
| Training time       | Minutes–hours | Hours–days            |
| Performance         | Good          | State-of-the-art      |

# Inside a Neural Network



## Neuron / Node

Receives weighted inputs, sums them, applies activation function, passes output forward



## Weights & Biases

Learnable parameters. Weights scale input importance; bias shifts the activation threshold



## Activation Functions

ReLU (most common), Sigmoid (0-1), Tanh (-1 to 1), Softmax (multi-class probabilities)



## Forward Pass

Data flows input→hidden→output, producing a prediction. One pass per training sample



## Loss Function

Measures prediction error. MSE for regression; Cross-Entropy for classification



## Backpropagation

Computes gradients of loss w.r.t. weights using chain rule, enabling weight updates

# Key Deep Learning Architectures

## CNN — Convolutional Neural Network

⚙ How: Convolutional filters detect local features — edges  
→ textures → objects

🚀 *Image classification · Object detection · Medical imaging  
· Autonomous vehicles*

## RNN / LSTM — Recurrent Neural Network

⚙ How: Hidden state carries memory across time steps;  
LSTM gates control information flow

🚀 *Machine translation · Speech recognition · Time series ·  
Sentiment analysis*

## ▫ Transformer — Attention-based Architecture

⚙ How: Self-attention weighs all token pairs simultaneously  
— no recurrence needed

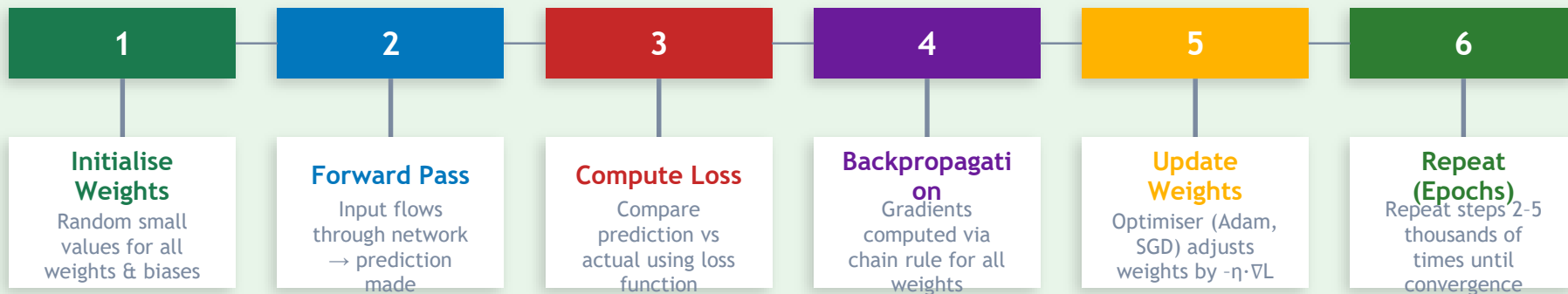
🚀 *GPT-4 · BERT · ChatGPT · DALL·E · Vision Transformers  
(ViT) · Code generation*

## GAN — Generative Adversarial Network

⚙ How: Generator vs Discriminator trained adversarially to  
create realistic data

🚀 *Image synthesis · Video generation · Data augmentation ·  
Deepfakes · Art*

# Training a Deep Learning Model



## Key Hyperparameters

- |                                             |                                                                         |                      |                                                                   |
|---------------------------------------------|-------------------------------------------------------------------------|----------------------|-------------------------------------------------------------------|
| • <b>Learning Rate (<math>\eta</math>):</b> | How big each weight update step is – too high diverges, too low is slow | • <b>Batch Size:</b> | Number of samples per gradient update – affects speed & stability |
| • <b>Epochs:</b>                            | Full passes over the training dataset – too many = overfitting          | • <b>Dropout:</b>    | Randomly zero neurons during training to prevent overfitting      |
| • <b>Optimiser:</b>                         | Adam (default), SGD, RMSProp – controls gradient descent strategy       |                      |                                                                   |

# Real-World Deep Learning Applications

## Computer Vision

Object detection · Face ID ·  
Medical scans · Self-driving cars ·  
Quality inspection

## Natural Language

ChatGPT · Google Translate · Text  
summarisation · Code generation ·  
Sentiment

## Healthcare

Cancer detection · Drug discovery ·  
Protein folding · Clinical notes ·  
Radiology

## Generative AI

DALL·E · Stable Diffusion ·  
MidJourney · Music generation ·  
Video synthesis

## Audio & Speech

Speech-to-text · Voice cloning ·  
Music composition · Speaker  
identification

## Science & Research

AlphaFold · Climate modelling ·  
Particle physics · Astronomy ·  
Materials

# Learning Roadmap & Career Paths in AI/ML/DL

## Phase 1

### Foundations

Weeks 1-4

- Python programming
- Linear Algebra & Calculus
- Probability & Statistics
- Data wrangling with Pandas

## Phase 2

### Classical ML

Weeks 5-10

- Scikit-learn algorithms
- Supervised & Unsupervised
- Model evaluation
- Feature engineering

## Phase 3

### Deep Learning

Weeks 11-18

- TensorFlow / PyTorch
- CNNs, RNNs, Transformers
- Training & optimisation
- Transfer learning

## Phase 4

### Specialise

Weeks 19+

- NLP / Vision / RL / GenAI
- Read research papers
- Build portfolio projects
- Kaggle competitions

Career Roles: ML Engineer • Data Scientist • AI Researcher • NLP Engineer • Computer Vision Engineer • MLOps

Engineer

# Key Takeaways

## AI

- Broadest field – simulating human intelligence
- Branches: ML, NLP, Vision, Robotics, GenAI
- Today: Narrow AI. Goal: AGI. Speculation: ASI
- Ethics matter: bias, privacy, jobs, safety

## ML

- Subset of AI – learns from data
- Types: Supervised · Unsupervised · Reinforcement
- Workflow: Collect → Clean → Engineer → Train → Evaluate → Deploy
- Tools: Python, Scikit-learn, Pandas, XGBoost

## DL

- Subset of ML – multi-layer neural networks
- No manual features – learns representations automatically
- Architectures: CNN, RNN, Transformer, GAN
- Powers ChatGPT, DALL·E, AlphaFold, Tesla FSD

Thank you! Keep Learning. The AI revolution is just beginning. 🚀